



Republic of Kenya — Competency Based Curriculum

GRADE 10 MATHEMATICS

LEARNER'S NOTES

Senior School — STEM Pathway



Aligned to the KICD Senior School Curriculum Design, Grade 10 Mathematics

For Learners aged 14–16 years

Strands: Numbers and Algebra • Measurements and Geometry • Statistics and Probability

About These Notes

Welcome, learner! These are your **Grade 10 Mathematics Notes**, written specially for you as you begin Senior School. This booklet has been developed based on the Kenya Institute of Curriculum Development (KICD) Senior School Curriculum Design for Grade 10 Mathematics. It covers the three strands you will study this year:

1. **Numbers and Algebra**
2. **Measurements and Geometry**
3. **Statistics and Probability**

Each topic in this book is organised the same way, so that you always know what to expect:

Learning Outcomes	What you should be able to do by the end of the topic.
Notes & Explanations	The main content, explained step by step with worked examples.
Key Idea boxes	Important facts, formulae and rules to remember.
Worked Examples	Fully solved problems showing you how to apply the ideas.
Think About It (Reflection)	Open questions to discuss with a partner or think about on your own.
Case Study	Real-life Kenyan situations that show how the mathematics is used.
Exercises	Practice questions to test your understanding as you go.
End of Topic Questions	A summary set of questions covering the whole topic.
Worksheets	Extra activity sheets at the end of each strand for more practice.

KEY IDEA

All the answers to the Exercises, End of Topic Questions and Worksheets in this book are found in the **Answer Key** chapter at the end of the book. Try each question fully on your own (or in your group) before checking the answer key.

Use a scientific calculator, mathematical tables, a ruler, a pair of compasses, a protractor and graph paper where needed. Work neatly, show all your steps, and always ask your teacher when you get stuck. Enjoy your Mathematics!

Table of Contents

STRAND 1: NUMBERS AND ALGEBRA

- 1.1 Real Numbers
- 1.2 Indices and Logarithms
- 1.3 Quadratic Expressions and Equations

STRAND 2: MEASUREMENTS AND GEOMETRY

- 2.1 Similarity and Enlargement
- 2.2 Reflection and Congruence
- 2.3 Rotation
- 2.4 Trigonometry I
- 2.5 Area of Polygons
- 2.6 Area of a Part of a Circle
- 2.7 Surface Area and Volume of Solids
- 2.8 Vectors I
- 2.9 Linear Motion

STRAND 3: STATISTICS AND PROBABILITY

- 3.1 Statistics I
- 3.2 Probability I

END MATTER

- Strand Worksheets
- Answer Key (all Exercises, End of Topic Questions and Worksheets)

STRAND 1.0: NUMBERS AND ALGEBRA

In this strand you will study Real Numbers, Indices and Logarithms, and Quadratic Expressions and Equations. These topics build your number sense and algebraic thinking, which you will use throughout Senior School Mathematics.

Topic 1.1: Real Numbers

By the end of this topic, you should be able to:

- classify whole numbers as odd, even, prime and composite;
- classify real numbers as rational and irrational;
- determine the reciprocal of real numbers by division, tables and calculators;
- apply reciprocals of real numbers in mathematical computations.

1.1.1 The Real Number System

Numbers surround us in everyday life — from counting animals at home to measuring distance to school. The set of **Real Numbers** ($\mathbb{R}$) contains all the numbers we use in Senior School Mathematics. Real numbers are divided into two main groups: **rational numbers** and **irrational numbers**.

KEY IDEA

Rational numbers can be written as a fraction a/b , where a and b are integers and $b \neq 0$. Examples: $1/2$, 3 , -4 , 0.75 , $0.333\dots$ (recurring).

Irrational numbers cannot be written as an exact fraction. Their decimal expansions are non-terminating and non-recurring. Examples: $\sqrt{2}$, $\sqrt{3}$, π .

Within the whole numbers, we classify numbers further:

Even number	A whole number divisible by 2 exactly, e.g. 2, 4, 6, 18, 100.
Odd number	A whole number not divisible by 2, e.g. 1, 3, 7, 15, 99.
Prime number	A whole number greater than 1 with exactly two factors: 1 and itself, e.g. 2, 3, 5, 7, 11, 13.
Composite number	A whole number greater than 1 with more than two factors, e.g. 4, 6, 8, 9, 12.

Worked Example 1

Classify the numbers 15, 17, 24 and $0.6767\dots$ as prime/composite and rational/irrational.

Solution:

$15 = 3 \times 5 \rightarrow$ composite (more than two factors)
 $17 \rightarrow$ only factors are 1 and 17 $\rightarrow$ prime
 $24 = 2^3 \times 3 \rightarrow$ composite
 $0.6767\dots$ is a recurring decimal, so it can be written as a fraction ($67/99$) $\rightarrow$ rational.

1.1.2 The Reciprocal of a Real Number

The **reciprocal** of a number is 1 divided by that number. If a number is x , its reciprocal is $1/x$ (provided $x \neq 0$). Multiplying a number by its reciprocal always gives 1.

KEY IDEA

Reciprocal of $x = 1 \div x$. The reciprocal of a fraction a/b is b/a (simply invert it). Zero has *no* reciprocal because division by zero is undefined.

Worked Example 2

Find the reciprocal of: (a) 8 (b) 0.4 (c) $2\frac{3}{4}$

Solution:

(a) Reciprocal of $8 = 1/8 = 0.125$
(b) Reciprocal of $0.4 = 1/0.4 = 10/4 = 2.5$
(c) $2\frac{3}{4} = 11/4$, so reciprocal $= 4/11 = 0.3636\dots$

Reciprocals may also be read directly from four-figure mathematical tables (the "Reciprocal" table), or found instantly by pressing the **1/x** (or x^{-1}) button on a scientific calculator.

THINK ABOUT IT

Why do you think zero has no reciprocal? Discuss with a partner what happens when you try to divide 1 by 0 on a calculator.

CASE STUDY: The Boda-Boda Fuel Tank

A boda-boda rider's fuel tank holds 4 litres and he uses $\frac{2}{5}$ of a litre for every kilometre travelled in town traffic. To find out how many kilometres he can ride per litre, he needs to work with the reciprocal of $\frac{2}{5}$, which is $\frac{5}{2} = 2.5$ km per litre. Understanding reciprocals therefore helps riders and small business owners plan fuel consumption and costs.

EXERCISE 1.1

1. Classify the following as odd, even, prime or composite: 21, 32, 41, 57, 64.
2. State whether each of the following is rational or irrational: 0.25, $\sqrt{5}$, 7, $\sqrt{16}$, 1.101001000...
3. Find the reciprocal of: (a) 12 (b) 0.25 (c) $1\frac{1}{5}$ (d) 0.125
4. A number and its reciprocal add up to 2. What is the number?
5. Using a calculator, find the reciprocal of 3.6, correct to 3 decimal places.
6. Explain, in your own words, why every whole number greater than 1 is either prime or composite.

End of Topic Questions - Topic 1.1

1. List all the prime numbers between 40 and 60.
2. Show that 0.4545... is a rational number by expressing it as a fraction.
3. Find the reciprocal of $5\frac{1}{3}$ and use it to work out $5\frac{1}{3} \times$ (its reciprocal).
4. A tank is filled at a rate of $\frac{3}{8}$ of a tank per hour. How many hours does it take to fill 1 whole tank? (Hint: use the reciprocal.)
5. Is the number 1 prime, composite, or neither? Justify your answer.

Topic 1.2: Indices and Logarithms

By the end of this topic, you should be able to:

- express numbers in index form and derive the laws of indices;
- apply the laws of indices in different situations;
- relate index notation to logarithm notation to base 10;
- determine and apply common logarithms in multiplication, division, powers and roots.

1.2.1 Indices

An **index** (plural: indices), also called a power or exponent, shows how many times a number (the base) is multiplied by itself. In a^n , a is the base and n is the index.

KEY IDEA - LAWS OF INDICES

1. $a^m \times a^n = a^{m+n}$
2. $a^m \div a^n = a^{m-n}$
3. $(a^m)^n = a^{mn}$
4. $a^0 = 1$ ($a \neq 0$)
5. $a^{-n} = 1/a^n$
6. $a^{1/n} = \sqrt[n]{a}$

Worked Example 1

Simplify: (a) $2^3 \times 2^2$ (b) $5^5 \div 5^2$ (c) $(3^2)^3$ (d) 4^{-2}

Solution:

- (a) $2^3 \times 2^2 = 2^5 = 32$
 (b) $5^5 \div 5^2 = 5^3 = 125$
 (c) $(3^2)^3 = 3^6 = 729$
 (d) $4^{-2} = 1/4^2 = 1/16$

1.2.2 Logarithms to Base 10

A **logarithm** is simply another way of writing an index statement. If $10^x = N$, then we say $x = \log_{10} N$ (read as "log to base 10 of N "). Since our number system is base 10, logarithms to base 10 are called **common logarithms**.

KEY IDEA - INDEX $\leftrightarrow$ LOGARITHM

$$10^x = N \leftrightarrow \log N = x$$

Laws of logarithms:

$$\log(A \times B) = \log A + \log B$$

$$\log(A \div B) = \log A - \log B$$

$$\log(A^n) = n \log A$$

Worked Example 2

Use logarithms to evaluate 23.4×5.62 (to 3 significant figures).

Solution:

$$\log 23.4 = 1.3692$$

$$\log 5.62 = 0.7497$$

$$\text{Sum} = 2.1189$$

$$\text{Antilog of } 2.1189 \approx 131.5$$

$$\text{So } 23.4 \times 5.62 \approx 131 \text{ (3 s.f.)}$$

Worked Example 3

Use logarithms to evaluate $\sqrt[3]{64.8}$ (the cube root of 64.8).

Solution:

$\log 64.8 = 1.8116$
Divide by 3: $1.8116 \div 3 = 0.6039$
Antilog of $0.6039 \approx 4.017$
So $\sqrt[3]{64.8} \approx 4.02$ (3 s.f.)

THINK ABOUT IT

Before calculators became common, workers such as surveyors and engineers relied heavily on logarithm tables to multiply and divide large numbers quickly. Why do you think logarithms make multiplication and division easier than working with the original numbers directly?

CASE STUDY: Measuring Earthquake Strength

Scientists use a logarithmic scale (the Richter scale) to measure the strength of earthquakes. An earthquake of magnitude 6 is 10 times stronger than one of magnitude 5, and 100 times stronger than one of magnitude 4. This is because each whole number increase represents a power of 10 — exactly how logarithms work! This shows how indices and logarithms are applied in real scientific measurement.

EXERCISE 1.2

1. Simplify: (a) $3^4 \times 3^2$ (b) $7^5 \div 7^3$ (c) $(2^3)^2$ (d) 6^{-1}
2. Write $1000 = 10^3$ in logarithmic form.
3. Write $\log_{10} 100\,000 = 5$ in index form.
4. Use logarithm tables or a calculator to evaluate 34.5×2.16 correct to 3 significant figures.
5. Use logarithms to evaluate $456.7 \div 12.3$.
6. Evaluate $\sqrt{28.9}$ using logarithms.
7. Simplify $2^x \times 2^3 = 2^5$, hence find x .

End of Topic Questions - Topic 1.2

1. Simplify $(5^2)^3 \div 5^4$.
2. Given that $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 6$ without using tables.
3. Use logarithms to evaluate $(23.6 \times 1.98) \div 4.05$, correct to 3 significant figures.
4. Solve for x : $3^x = 81$.
5. A population of bacteria doubles every hour, starting with 2 bacteria. Write an expression in index form for the number of bacteria after n hours, and find the number after 5 hours.

Topic 1.3: Quadratic Expressions and Equations

By the end of this topic, you should be able to:

- form quadratic expressions from different situations;
- derive and apply quadratic identities;
- factorise quadratic expressions;
- form and solve quadratic equations by factorisation, and apply them to real life.

1.3.1 Quadratic Expressions

A **quadratic expression** is an algebraic expression in which the highest power of the unknown is 2. It has the general form $ax^2 + bx + c$, where a , b and c are constants and $a \neq 0$.

KEY IDEA - QUADRATIC IDENTITIES

$$(a + b)^2 = a^2 + 2ab + b^2$$

$$(a - b)^2 = a^2 - 2ab + b^2$$

$$(a + b)(a - b) = a^2 - b^2 \text{ (difference of two squares)}$$

Worked Example 1

Expand $(x + 5)^2$ and factorise $x^2 - 49$.

Solution:

$$(x + 5)^2 = x^2 + 10x + 25$$

$$x^2 - 49 = x^2 - 7^2 = (x + 7)(x - 7)$$

1.3.2 Factorisation and Solving Quadratic Equations

A quadratic equation is a quadratic expression set equal to zero, e.g. $x^2 - 5x + 6 = 0$. To solve by **factorisation**, find two numbers that multiply to give the product ac and add to give b , then factor and use the zero-product rule: if $PQ = 0$, then $P = 0$ or $Q = 0$.

Worked Example 2

Solve $x^2 - 5x + 6 = 0$

Solution:

We need two numbers that multiply to give 6 and add to give -5 : these are -2 and -3 .

$$x^2 - 2x - 3x + 6 = 0$$

$$x(x - 2) - 3(x - 2) = 0$$

$$(x - 2)(x - 3) = 0$$

$$x = 2 \text{ or } x = 3$$

Worked Example 3 - Real-life application

A rectangular plot has a length that is 3 m more than its width. If the area of the plot is 108 m^2 , find its dimensions.

Solution:

Let width = x m, then length = $(x + 3)$ m.

$$x(x + 3) = 108$$

$$x^2 + 3x - 108 = 0$$

$$(x + 12)(x - 9) = 0$$

$$x = -12 \text{ (rejected, since length cannot be negative) or } x = 9$$

$$\text{Width} = 9 \text{ m, Length} = 12 \text{ m}$$

THINK ABOUT IT

In Worked Example 3, we rejected the negative solution $x = -12$. Can you think of another real-life quadratic problem where a negative solution would need to be rejected? Discuss with your classmates.

CASE STUDY: Designing a School Kitchen Garden

A school wants to build a rectangular kitchen garden along one wall of the compound, using 40 m of fencing wire for the three open sides. If the area enclosed is to be 150 m^2 , the students can form and solve a quadratic equation to find the possible dimensions of the garden, helping the school plan its Grade 10 Agriculture and Home Science projects.

EXERCISE 1.3

1. Expand: (a) $(x + 4)^2$ (b) $(2x - 3)^2$ (c) $(x + 6)(x - 6)$
2. Factorise: (a) $x^2 - 9$ (b) $x^2 + 7x + 12$ (c) $x^2 - 3x - 10$
3. Solve the following quadratic equations by factorisation: (a) $x^2 - 7x + 12 = 0$ (b) $x^2 + 2x - 15 = 0$ (c) $2x^2 - 5x - 3 = 0$
4. The length of a rectangle is 4 cm more than its width. The area is 45 cm^2 . Find the dimensions of the rectangle.
5. Form a quadratic equation whose roots are 4 and -3 .

End of Topic Questions - Topic 1.3

1. Use the identity $(a - b)^2 = a^2 - 2ab + b^2$ to evaluate 98^2 without a calculator.
2. Factorise completely: $3x^2 - 12$.
3. Solve: $x^2 - x - 6 = 0$.
4. A ball is thrown and its height h (in metres) after t seconds is given by $h = 20t - 5t^2$. Find the values of t when the ball is at ground level ($h = 0$).
5. Two consecutive positive integers have a product of 90. Find the integers.

STRAND 2.0: MEASUREMENTS AND GEOMETRY

This strand covers transformations (similarity, reflection, rotation), trigonometry, area, surface area and volume, vectors, and linear motion — the tools you need to describe and measure our physical world.

Topic 2.1: Similarity and Enlargement

By the end of this topic, you should be able to:

- determine the centre of enlargement and the linear scale factor for similar figures;
- construct the image of an object under an enlargement given the centre and scale factor;
- determine the area and volume scale factors of similar figures and solids;
- apply similarity and enlargement to real-life situations.

2.1.1 Similar Figures and Enlargement

Two figures are **similar** if they have the same shape but not necessarily the same size — their corresponding angles are equal and corresponding sides are in the same ratio. An **enlargement** is a transformation that produces similar figures using a fixed point (the **centre of enlargement**) and a **linear scale factor (L.S.F)**.

KEY IDEA

Linear Scale Factor (L.S.F) = (length on image) $\div$ (corresponding length on object)

Area Scale Factor (A.S.F) = (L.S.F)²

Volume Scale Factor (V.S.F) = (L.S.F)³

Worked Example 1

Triangle ABC has sides 3 cm, 4 cm and 5 cm. It is enlarged to triangle A'B'C' with corresponding sides 9 cm, 12 cm and 15 cm. Find the linear, area and volume scale factors.

Solution:

$$\text{L.S.F} = 9/3 = 3$$

$$\text{A.S.F} = 3^2 = 9$$

$$\text{V.S.F} = 3^3 = 27 \text{ (used only if these were 3-D solids)}$$

Worked Example 2

A model car is 12 cm long and its actual size is 4.8 m long. Find the linear scale factor and the ratio of their volumes.

Solution:

$$4.8 \text{ m} = 480 \text{ cm. L.S.F (model : actual)} = 12 : 480 = 1 : 40$$

$$\text{V.S.F} = 1^3 : 40^3 = 1 : 64\,000$$

THINK ABOUT IT

Photocopy machines can enlarge or reduce documents by a given percentage. If you enlarge an A4 page by 150%, what is the linear scale factor, and by what factor does the area of the page increase?

CASE STUDY: Architectural Scale Models

Architects build small scale models of buildings before construction begins. A model of a school building might use a scale of 1:100. This means every 1 cm on the model represents 100 cm (1 m) on the real building. Enlargement and scale factors help architects, surveyors, and engineers plan accurately.

EXERCISE 2.1

1. Two similar rectangles have lengths 5 cm and 15 cm. Find the linear scale factor and the ratio of their areas.
2. A cube of side 2 cm is enlarged to a cube of side 6 cm. Find the L.S.F, A.S.F and V.S.F.
3. A map is drawn using a scale of 1:50 000. If a road measures 8 cm on the map, find its actual length in kilometres.
4. Two similar cylinders have volumes 64 cm³ and 216 cm³. Find the ratio of their heights.
5. Describe, using your own example, how you would find the centre of enlargement given an object and its image.

1. Two similar triangles have areas 18 cm^2 and 50 cm^2 . Find the ratio of their corresponding sides.
2. A solid has a volume of 27 cm^3 . A similar solid has a linear scale factor of 2 relative to it. Find the volume of the larger solid.
3. A photograph measuring 6 cm by 9 cm is enlarged so that its longer side becomes 27 cm. Find the new width and the area scale factor.

Topic 2.2: Reflection and Congruence

By the end of this topic, you should be able to:

- identify lines of symmetry in plane figures and determine properties of reflection;
- draw an image given an object and a mirror line on a plane surface and Cartesian plane;
- determine the equation of a mirror line given an object and its image;
- carry out congruence tests for triangles.

2.2.1 Reflection

A **reflection** is a transformation that produces a mirror image of an object across a straight line called the **mirror line (line of symmetry)**. Every point on the image is the same perpendicular distance from the mirror line as the corresponding point on the object.

KEY IDEA - PROPERTIES OF REFLECTION

1. The object and image are congruent (same size and shape).
2. The mirror line is the perpendicular bisector of the line joining any point to its image.
3. Reflection in the line $y = x$ swaps the coordinates: $(a, b) \rightarrow (b, a)$.
4. Reflection in the x-axis: $(a, b) \rightarrow (a, -b)$. Reflection in the y-axis: $(a, b) \rightarrow (-a, b)$.

Worked Example 1

Find the image of point $P(2, 3)$ under reflection in the y-axis, then in the line $y = x$.

Solution:

Reflection in y-axis: $P(2, 3) \rightarrow P'(-2, 3)$

Reflection in $y = x$: $P(2, 3) \rightarrow P''(3, 2)$

2.2.2 Congruence

Two figures are **congruent** if they are identical in shape and size. For triangles, we can prove congruence using one of the following tests:

KEY IDEA - CONGRUENCE TESTS FOR TRIANGLES

SSS – three sides equal

SAS – two sides and the included angle equal

ASA / AAS – two angles and a side equal

RHS – right angle, hypotenuse and one side equal

THINK ABOUT IT

Look at your own face in a mirror. Is your face perfectly symmetrical? Discuss how reflection appears in nature (e.g. butterfly wings, leaves) and in Kenyan traditional art and architecture.

CASE STUDY: Kanga and Kitenge Patterns

Many Kenyan kanga and kitenge fabric designs use reflective symmetry to create beautiful, balanced patterns. Designers often draw one half of a pattern and reflect it across a central line to complete the design — a direct real-life application of the mathematics of reflection.

EXERCISE 2.2

1. Draw triangle $A(1,1)$, $B(4,1)$, $C(4,3)$ on a Cartesian plane and find its image under reflection in the x-axis.
2. State the coordinates of the image of point $(5, -2)$ after reflection in the line $y = x$.
3. Find the equation of the mirror line if point $(2, 1)$ maps onto $(2, 5)$ under reflection.
4. State which congruence test (SSS, SAS, ASA, RHS) would you use to show two right-angled triangles with equal hypotenuses and one equal side are congruent.
5. Draw a square and state the number of lines of symmetry it has.

End of Topic Questions - Topic 2.2

1. Reflect the point $(-3, 4)$ in the y-axis, then in the x-axis. Give both images.
2. An isosceles triangle has how many lines of symmetry? Sketch and show them.
3. Two triangles have sides 5 cm, 6 cm, 7cm and 5 cm, 6cm, 7cm respectively. State the congruence test that proves they are congruent.

Topic 2.3: Rotation

By the end of this topic, you should be able to:

- determine the properties of rotation;
- rotate an object given the centre and angle of rotation;
- determine the centre and angle of rotation given an object and its image;
- determine the order of rotational symmetry of plane figures and solids.

2.3.1 Rotation

A **rotation** is a transformation that turns a figure about a fixed point called the **centre of rotation**, through a given **angle of rotation**, in a clockwise or anticlockwise direction.

KEY IDEA - PROPERTIES OF ROTATION

1. The object and its image are congruent.
2. Every point moves through the same angle about the centre.
3. The distance from the centre of rotation to any point equals the distance from the centre to its image.

KEY IDEA - COMMON ROTATION RULES (about the origin)

90° anticlockwise: $(x, y) \rightarrow (-y, x)$

180°: $(x, y) \rightarrow (-x, -y)$

270° anticlockwise (90° clockwise): $(x, y) \rightarrow (y, -x)$

Worked Example 1

Find the image of point A(3, 2) after a rotation of 90° anticlockwise about the origin.

Solution:

Using $(x, y) \rightarrow (-y, x)$: $A(3, 2) \rightarrow A'(-2, 3)$

2.3.2 Rotational Symmetry

A plane figure has **rotational symmetry** if it looks the same after being rotated through an angle less than 360° about its centre. The **order of rotational symmetry** is the number of times the figure fits onto itself during one complete turn.

Worked Example 2

State the order of rotational symmetry of a regular pentagon.

Solution:

A regular pentagon has 5 equal sides and fits onto itself 5 times in a full turn. Order of rotational symmetry = 5.

THINK ABOUT IT

Look at the wheel of a bicycle or a car. What is the order of rotational symmetry of a wheel with, say, 5 spokes evenly spaced? How does this relate to how smoothly the wheel spins?

CASE STUDY: The Clock Hands

The minute hand of a clock rotates 360° about the centre of the clock face every 60 minutes, while the hour hand rotates 360° every 12 hours. Engineers use the mathematics of rotation and angular speed to design accurate clocks, gears, and machine parts.

EXERCISE 2.3

1. Find the image of B(−2, 5) under a rotation of 180° about the origin.
2. Find the image of C(4, −1) under a rotation of 90° clockwise about the origin.
3. State the order of rotational symmetry of: (a) an equilateral triangle (b) a regular hexagon (c) a rectangle (non-square).
4. A point P(1, 4) is rotated about the origin and maps onto P'(−4, 1). Determine the angle and direction of rotation.

5. Give the order of rotational symmetry of a circle.

End of Topic Questions - Topic 2.3

1. Rotate the point (5, 0) through 90° anticlockwise about the origin, then rotate the result through a further 90° anticlockwise. State the final coordinates.
2. A regular octagon is rotated about its centre. What is its order of rotational symmetry?
3. Explain the difference between reflection and rotation using a real-life example for each.

Topic 2.4: Trigonometry I

By the end of this topic, you should be able to:

- determine trigonometric ratios of acute angles from tables and calculators;
- relate sines and cosines of complementary angles;
- determine trigonometric ratios of special angles 30°, 45°, 60° and 90°;
- apply trigonometric ratios to angles of elevation and depression.

2.4.1 Trigonometric Ratios

In a right-angled triangle, the trigonometric ratios relate the angles to the ratio of the sides. For an acute angle θ , using the opposite side, adjacent side and hypotenuse:

KEY IDEA - SOH CAH TOA

$\sin \theta = \text{Opposite} / \text{Hypotenuse}$
 $\cos \theta = \text{Adjacent} / \text{Hypotenuse}$
 $\tan \theta = \text{Opposite} / \text{Adjacent}$

2.4.2 Complementary Angles and Special Angles

KEY IDEA

Two angles are complementary if they add up to 90°. For any angle θ : $\sin \theta = \cos(90^\circ - \theta)$ and $\cos \theta = \sin(90^\circ - \theta)$.

θ	30°	45°	60°	90°
$\sin \theta$	1/2	1/√2	√3/2	1
$\cos \theta$	√3/2	1/√2	1/2	0
$\tan \theta$	1/√3	1	√3	undefined

Worked Example 1

A ladder 5 m long leans against a wall, making an angle of 60° with the ground. Find how high up the wall the ladder reaches.

Solution:

Height = $5 \times \sin 60^\circ = 5 \times 0.8660 = 4.33 \text{ m}$

2.4.3 Angles of Elevation and Depression

The **angle of elevation** is the angle measured upwards from the horizontal to an object above. The **angle of depression** is the angle measured downwards from the horizontal to an object below.

Worked Example 2

From the top of a cliff 60 m high, the angle of depression of a boat is 25°. Find the horizontal distance of the boat from the base of the cliff.

Solution:

$\tan 25^\circ = 60 / d \Rightarrow d = 60 / \tan 25^\circ = 60 / 0.4663 \approx 128.7 \text{ m}$

THINK ABOUT IT

A pilot flying an aircraft views the runway at an angle of depression. A person on the ground looking up at the same aircraft sees it at the angle of elevation. Explain why these two angles are always equal.

CASE STUDY: Measuring the Height of Mount Kenya

Surveyors cannot always climb to the top of a mountain to measure its height directly. Instead, they stand a known distance away and measure the angle of elevation to the peak using a theodolite (an instrument based on trigonometry). Using tangent ratios, they can then calculate the mountain's height accurately.

EXERCISE 2.4

1. Find $\sin 42^\circ$, $\cos 42^\circ$ and $\tan 42^\circ$ using a calculator, correct to 4 decimal places.
2. A tree casts a shadow of 12 m when the angle of elevation of the sun is 35° . Find the height of the tree.
3. Use the special angle values to evaluate $\sin 30^\circ + \cos 60^\circ$.
4. From a window 15 m above the ground, the angle of depression to a parked car is 20° . Find the distance of the car from the base of the building.
5. Show that $\sin 50^\circ = \cos 40^\circ$, using the complementary angle relationship.

End of Topic Questions - Topic 2.4

1. A ladder leaning against a wall makes an angle of 70° with the ground. If the foot of the ladder is 2 m from the wall, find the length of the ladder.
2. The angle of elevation of the top of a flagpole from a point 20 m away is 40° . Find the height of the flagpole.
3. Evaluate, without a calculator: $\tan 45^\circ + \sin 30^\circ \times \cos 60^\circ$.

Topic 2.5: Area of Polygons

By the end of this topic, you should be able to:

- derive and apply the formula $\text{Area} = \frac{1}{2}ab \sin C$ for a triangle;
- determine the area of a triangle using Heron's formula;
- determine the area of quadrilaterals, regular and irregular polygons.

2.5.1 Area of a Triangle

KEY IDEA

Given two sides and the included angle: $\text{Area} = \frac{1}{2} ab \sin C$

Heron's Formula (given three sides a, b, c):

$s = (a + b + c) / 2$ (the semi-perimeter)

$\text{Area} = \sqrt{[s(s-a)(s-b)(s-c)]}$

Worked Example 1

Find the area of a triangle with sides 8 cm and 10 cm, with an included angle of 50° .

Solution:

$\text{Area} = \frac{1}{2} \times 8 \times 10 \times \sin 50^\circ = 40 \times 0.766 = 30.6 \text{ cm}^2$

Worked Example 2 - Heron's Formula

Find the area of a triangle with sides 7 cm, 8 cm and 9 cm.

Solution:

$s = (7+8+9)/2 = 12$

$\text{Area} = \sqrt{[12(12-7)(12-8)(12-9)]} = \sqrt{[12 \times 5 \times 4 \times 3]} = \sqrt{720} \approx 26.83 \text{ cm}^2$

2.5.2 Area of Quadrilaterals and Regular Polygons

Common quadrilateral area formulae: rectangle = $l \times w$; parallelogram = base $\times$ height; trapezium = $\frac{1}{2}(a+b)h$; rhombus/kite = $\frac{1}{2} \times d_1 \times d_2$ (diagonals).

KEY IDEA

A regular polygon with n sides can be divided into n congruent isosceles triangles from its centre. The area of the polygon is n times the area of one triangle, where each triangle's angle at the centre is $360^\circ/n$.

THINK ABOUT IT

Football pitches, basketball courts and volleyball courts are all rectangles, but a football is often made of pentagons and hexagons stitched together. Why might sports equipment designers choose regular polygons for some designs?

CASE STUDY: Tiling a Compound

A homeowner wants to tile a triangular section of a compound using interlocking paving blocks shaped as regular hexagons. Knowing how to calculate the area of these polygons helps the contractor estimate the exact number of paving blocks and the total cost of materials needed for the job.

EXERCISE 2.5

1. Find the area of a triangle with sides 6 cm and 9 cm and an included angle of 65° .
2. Use Heron's formula to find the area of a triangle with sides 5 cm, 6 cm and 7 cm.
3. Find the area of a trapezium with parallel sides 8 cm and 12 cm, and height 5 cm.
4. Find the area of a regular hexagon with side length 4 cm. (Hint: divide it into 6 equilateral triangles.)
5. A kite has diagonals 10 cm and 14 cm. Find its area.

End of Topic Questions - Topic 2.5

1. Find the area of a parallelogram with sides 9 cm and 6 cm, with an included angle of 75° .
2. A triangular garden has sides 15 m, 18 m and 24 m. Use Heron's formula to find its area.
3. Find the area of a regular octagon with side length 3 cm, correct to 1 decimal place.

Topic 2.6: Area of a Part of a Circle

By the end of this topic, you should be able to:

- determine the area of an annulus, a sector, an annular sector, and a segment of a circle;
- determine the area of the common region between two intersecting circles;
- apply the area of a part of a circle to real-life situations.

2.6.1 Sectors, Annuli and Segments

KEY IDEA

Area of a circle = πr^2

Area of a sector = $(\theta/360) \times \pi r^2$, where θ is the angle at the centre

Area of an **annulus** (ring between two concentric circles) = $\pi(R^2 - r^2)$, where R is the outer radius and r is the inner radius

Area of an **annular sector** = $(\theta/360) \times \pi(R^2 - r^2)$

Area of a **segment** = Area of sector – Area of triangle = $(\theta/360)\pi r^2 - \frac{1}{2}r^2 \sin \theta$

Worked Example 1

Find the area of a sector of a circle of radius 14 cm, subtending an angle of 60° at the centre. (Take $\pi = 22/7$)

Solution:

$$\text{Area} = (60/360) \times 22/7 \times 14^2 = (1/6) \times 22/7 \times 196 = 102.67 \text{ cm}^2$$

Worked Example 2

A circular dartboard has an outer radius of 20 cm and an inner (bullseye) radius of 5 cm. Find the area of the annulus between them. (Take $\pi = 3.142$)

Solution:

$$\text{Area} = \pi(20^2 - 5^2) = 3.142 \times (400 - 25) = 3.142 \times 375 = 1178.25 \text{ cm}^2$$

THINK ABOUT IT

A car's windscreen wiper sweeps out a sector shape as it cleans the glass. If you know the length of the wiper blade and the angle it sweeps, how would you calculate the area of glass it cleans in one sweep?

CASE STUDY: Designing a Roundabout Garden

County governments often plant flowers and grass in the circular islands (roundabouts) found at road junctions. When part of the roundabout is paved for pedestrian crossing and the rest is planted, engineers use the area of sectors and segments to calculate how much soil, grass or paving material is required for each section.

EXERCISE 2.6

1. Find the area of a sector of radius 21 cm and angle 90° . (Take $\pi = 22/7$)
2. An annulus has an outer radius of 10 cm and inner radius of 6 cm. Find its area. (Take $\pi = 3.142$)
3. Find the area of a segment cut off by a chord in a circle of radius 8 cm, if the angle at the centre is 100° .
4. An annular sector has outer radius 12 cm, inner radius 8 cm and angle 45° . Find its area.
5. Two circles of radius 6 cm each intersect such that the distance between their centres is 6 cm. Describe, in words, how you would find the area of the overlapping (common) region.

End of Topic Questions - Topic 2.6

1. Find the area of a sector with radius 10.5 cm and angle 150° .
2. A car wiper blade of length 40 cm sweeps through an angle of 120° . Find the area it cleans.
3. Find the area of a segment of a circle of radius 9 cm, where the angle subtended at the centre is 80° .

Topic 2.7: Surface Area and Volume of Solids

By the end of this topic, you should be able to:

- determine the surface area of prisms, pyramids, cones, frustums and spheres;
- calculate the surface area and volume of composite solids;
- determine the volume of prisms, pyramids, cones, frustums and spheres.

2.7.1 Surface Area and Volume Formulae

KEY IDEA - VOLUME FORMULAE

Prism: $V = \text{Area of cross-section} \times \text{length}$

Pyramid: $V = \frac{1}{3} \times \text{base area} \times \text{height}$

Cone: $V = \frac{1}{3}\pi r^2 h$

Sphere: $V = \frac{4}{3}\pi r^3$

Frustum: $V = \text{Volume of full cone} - \text{Volume of small cone removed}$

KEY IDEA - SURFACE AREA FORMULAE

Cylinder (closed): $S.A = 2\pi rh + 2\pi r^2$

Cone (curved + base): $S.A = \pi rl + \pi r^2$ (l = slant height)

Sphere: $S.A = 4\pi r^2$

Worked Example 1

Find the volume of a cone with base radius 6 cm and height 10 cm. (Take $\pi = 3.142$)

Solution:

$$V = \frac{1}{3} \times 3.142 \times 6^2 \times 10 = \frac{1}{3} \times 3.142 \times 36 \times 10 = 377.04 \text{ cm}^3$$

Worked Example 2

A closed cylindrical tank has a radius of 3.5 m and height of 5 m. Find its total surface area. (Take $\pi = 22/7$)

Solution:

$$\begin{aligned} S.A &= 2\pi rh + 2\pi r^2 = 2(22/7)(3.5)(5) + 2(22/7)(3.5)^2 \\ &= 110 + 77 = 187 \text{ m}^2 \end{aligned}$$

Worked Example 3 - Composite Solid

A solid is made up of a cylinder of radius 4 cm and height 10 cm, topped with a hemisphere of radius 4 cm. Find its total volume. (Take $\pi = 3.142$)

Solution:

$$\text{Volume of cylinder} = \pi r^2 h = 3.142 \times 16 \times 10 = 502.72 \text{ cm}^3$$

$$\text{Volume of hemisphere} = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3} \times 3.142 \times 64 = 134.06 \text{ cm}^3$$

$$\text{Total volume} = 502.72 + 134.06 = 636.78 \text{ cm}^3$$

THINK ABOUT IT

Why do you think water tanks are often made in the shape of a cylinder rather than a cube? Consider surface area, cost of material, and strength in your answer.

CASE STUDY: Building a Grain Silo

Farmers in Kenya often store maize in cylindrical silos topped with a conical roof. Calculating the volume of the cylinder and cone separately allows the farmer to know exactly how many bags of maize the silo can hold, while calculating the surface area helps in estimating the amount of metal sheeting needed to construct it.

EXERCISE 2.7

1. Find the volume of a triangular prism whose cross-sectional area is 24 cm^2 and length is 15 cm.

2. Find the volume of a sphere of radius 7 cm. (Take $\pi = 22/7$)
3. Find the total surface area of a closed cone with base radius 5 cm and slant height 13 cm.
4. A frustum is formed by cutting off the top of a cone of height 12 cm and base radius 9 cm, at a height of 4 cm from the apex where the radius is 3 cm. Find the volume of the frustum.
5. A solid is composed of a cube of side 6 cm with a square pyramid of height 4 cm on top. Find the total volume.

End of Topic Questions - Topic 2.7

1. Find the surface area of a sphere of radius 10.5 cm.
2. A cylindrical tank of radius 2.1 m and height 3 m is to be filled with water. Find its capacity in litres ($1 \text{ m}^3 = 1000 \text{ litres}$).
3. Find the volume of a square-based pyramid with base side 8 cm and height 15 cm.

Topic 2.8: Vectors I

By the end of this topic, you should be able to:

- explain vector and scalar quantities, and represent vectors geometrically;
- add vectors and multiply vectors by a scalar;
- determine column vectors, position vectors, magnitude and midpoint of a vector;
- determine a translation vector as a transformation.

2.8.1 Vectors and Scalars

A **scalar quantity** has magnitude (size) only, e.g. mass, temperature, time. A **vector quantity** has both magnitude and direction, e.g. displacement, velocity, force. A vector is represented by a directed line segment and written in bold, e.g. **a**, or with an arrow, e.g. **AB**.

KEY IDEA - COLUMN VECTORS

A vector from point A(x_1, y_1) to B(x_2, y_2) is written as a column vector: $\vec{AB} = (x_2 - x_1, y_2 - y_1)$, displayed as a column with the x-component on top and y-component below.

KEY IDEA - OPERATIONS ON VECTORS

Addition: $(a, b) + (c, d) = (a+c, b+d)$ [add corresponding components]

Scalar multiplication: $k(a, b) = (ka, kb)$

Magnitude: $|(a, b)| = \sqrt{a^2 + b^2}$

Midpoint of AB: $M = ((x_1+x_2)/2, (y_1+y_2)/2)$

Worked Example 1

Given A(2, 3) and B(6, 7), find (a) the column vector $\vec{AB}$, (b) its magnitude, (c) the midpoint of AB.

Solution:

$$(a) \vec{AB} = (6-2, 7-3) = (4, 4)$$

$$(b) |\vec{AB}| = \sqrt{4^2+4^2} = \sqrt{32} = 5.66 \text{ (3 s.f.)}$$

$$(c) \text{Midpoint} = ((2+6)/2, (3+7)/2) = (4, 5)$$

Worked Example 2

Given $\mathbf{p} = (3, -1)$ and $\mathbf{q} = (-2, 5)$, find $2\mathbf{p} + \mathbf{q}$.

Solution:

$$2\mathbf{p} = (6, -2)$$

$$2\mathbf{p} + \mathbf{q} = (6-2, -2+5) = (4, 3)$$

THINK ABOUT IT

A pilot wants to fly due north, but a strong wind is blowing from the west. Explain, using the idea of vectors, why the plane might actually end up travelling in a slightly different direction than intended.

CASE STUDY: GPS Navigation

GPS systems in vehicles and smartphones use vectors to calculate the shortest route and direction from your current position to a destination. Displacement vectors combine distance and direction, allowing navigation apps to guide drivers efficiently across Kenyan roads and cities.

EXERCISE 2.8

1. Given P(1, 2) and Q(5, 8), find the column vector $\vec{PQ}$ and its magnitude.
2. If $\mathbf{a} = (3, 4)$ and $\mathbf{b} = (-1, 2)$, find $\mathbf{a} + \mathbf{b}$ and $\mathbf{a} - \mathbf{b}$.
3. Find the midpoint of the line joining $(-3, 4)$ and $(7, -2)$.
4. Find the magnitude of the vector $(5, 12)$.
5. Given $\mathbf{u} = (2, -3)$, find $3\mathbf{u}$.

End of Topic Questions - Topic 2.8

1. A translation moves point $(2, 3)$ to $(7, -1)$. Write down the translation vector.
2. Given $\mathbf{a} = (4, -2)$ and $\mathbf{b} = (1, 5)$, find $|2\mathbf{a} + \mathbf{b}|$.
3. The position vectors of A and B relative to the origin are $(2,1)$ and $(6,9)$ respectively. Find $\vec{AB}$ and the distance AB.

Topic 2.9: Linear Motion

By the end of this topic, you should be able to:

- explain distance, displacement, speed, velocity and acceleration;
- draw and interpret displacement-time and velocity-time graphs;
- determine the relative speed of two moving bodies.

2.9.1 Key Terms in Linear Motion

Distance	Total length of the path travelled (scalar quantity, in metres/kilometres).
Displacement	Distance moved in a specified direction (vector quantity).
Speed	Distance travelled per unit time = Distance ÷ Time.
Velocity	Displacement per unit time (speed in a given direction).
Acceleration	Rate of change of velocity = (Final velocity – Initial velocity) ÷ Time.

Worked Example 1

A car accelerates from 10 m/s to 25 m/s in 5 seconds. Find its acceleration.

Solution:

$$\text{Acceleration} = (25 - 10) / 5 = 15/5 = 3 \text{ m/s}^2$$

2.9.2 Displacement-Time and Velocity-Time Graphs

KEY IDEA

- On a **displacement-time graph**, the gradient (slope) at any point gives the velocity.
- On a **velocity-time graph**, the gradient gives the acceleration, and the area under the graph gives the distance travelled.

2.9.3 Relative Speed

KEY IDEA

- When two bodies move in the **same direction**, relative speed = difference of their speeds.
- When two bodies move in **opposite directions**, relative speed = sum of their speeds.

Worked Example 2

Two matatus leave the same bus stage at the same time, one travelling at 60 km/h and the other at 80 km/h in the same direction. Find their relative speed, and how far apart they will be after 2 hours.

Solution:

$$\begin{aligned} \text{Relative speed} &= 80 - 60 = 20 \text{ km/h} \\ \text{Distance apart after 2 hours} &= 20 \times 2 = 40 \text{ km} \end{aligned}$$

THINK ABOUT IT

When you sit in a moving matatu and another vehicle overtakes you slowly, it can seem to be almost standing still relative to you, even though both vehicles are moving fast relative to the ground. Explain why, using the idea of relative speed.

CASE STUDY: Road Safety and Stopping Distance

Traffic police and driving schools use the concepts of speed, velocity and acceleration to teach drivers about safe stopping distances. A vehicle travelling faster needs a greater distance (and time) to decelerate to a stop, which is why speed limits are stricter in busy town centres and near schools.

EXERCISE 2.9

1. A cyclist covers 15 km in 45 minutes. Find her average speed in km/h.

2. A car decelerates from 30 m/s to rest in 6 seconds. Find its acceleration (state the sign).
3. Two buses are 300 km apart and travel towards each other at 50 km/h and 70 km/h. Find the time they take to meet.
4. Sketch a velocity-time graph for a car that accelerates uniformly from 0 to 20 m/s in 10 seconds, then travels at constant velocity for 15 seconds. Find the total distance covered.
5. Explain the difference between distance and displacement using an example of a learner walking around a rectangular school field and back to the starting point.

End of Topic Questions - Topic 2.9

1. A train travels 240 km in 3 hours. Find its average speed.
2. Two cars travel in the same direction at 90 km/h and 110 km/h. Find their relative speed.
3. A body starts from rest and accelerates uniformly at 2 m/s^2 for 8 seconds. Find its final velocity and the distance covered.

STRAND 3.0: STATISTICS AND PROBABILITY

This strand introduces you to collecting, organising, representing and interpreting data, as well as understanding chance and probability — essential skills for research, business, and everyday decision-making.

Topic 3.1: Statistics I

By the end of this topic, you should be able to:

- collect data from real-life sources and draw frequency distribution tables;
- determine the mean, mode and median of grouped and ungrouped data;
- represent and interpret data using histograms and frequency polygons.

3.1.1 Collecting and Organising Data

Data refers to facts or information collected for analysis. Data can be organised into a **frequency distribution table**, which shows how often each value (or group of values) occurs.

Worked Example 1 - Ungrouped Data

The number of siblings of 10 learners are: 2, 1, 3, 2, 2, 0, 1, 3, 2, 1. Draw a frequency distribution table.

Solution:

Number of siblings	0	1	2	3
Frequency	1	3	4	2

3.1.2 Mean, Mode and Median

KEY IDEA - UNGROUPED DATA

- Mean = (Sum of all values) ÷ (Number of values)
- Mode = the value that occurs most frequently
- Median = the middle value when data is arranged in order (average of the two middle values if n is even)

KEY IDEA - GROUPED DATA

- Mean = $\Sigma fx \div \Sigma f$ (x = midpoint of each class, f = frequency)
- Modal class = the class with the highest frequency
- Median class = the class containing the middle value (locate using cumulative frequency)

Worked Example 2

Using the data in Worked Example 1, find the mean, mode and median.

Solution:

- Mean = $(2+1+3+2+2+0+1+3+2+1)/10 = 17/10 = 1.7$
- Mode = 2 (occurs 4 times)
- Arranged in order: 0,1,1,1,2,2,2,2,3,3 → median = $(2+2)/2 = 2$ (average of 5th and 6th values)

Worked Example 3 - Grouped Data

Find the mean of the following distribution of marks:

Marks	1-10	11-20	21-30	31-40
Frequency	3	7	6	4

Solution:

- Midpoints (x): 5.5, 15.5, 25.5, 35.5
- $\Sigma fx = (3 \times 5.5) + (7 \times 15.5) + (6 \times 25.5) + (4 \times 35.5) = 16.5 + 108.5 + 153 + 142 = 420$
- $\Sigma f = 3 + 7 + 6 + 4 = 20$
- Mean = $420/20 = 21$

3.1.3 Histograms and Frequency Polygons

A **histogram** represents grouped data using bars with no gaps between them (bar width represents the class interval, and bar

height represents frequency, or frequency density for unequal class widths). A **frequency polygon** is drawn by joining the midpoints of the tops of histogram bars with straight lines.

THINK ABOUT IT

A newspaper reports that "the average household in a certain estate has 4.3 children." Since you cannot have a fraction of a child, what does this "average" (mean) actually represent, and why is it still a useful statistic?

CASE STUDY: County Health Data

County health officials collect data on the ages of patients visiting a clinic each month. By organising this data into frequency tables and drawing histograms, they can identify which age groups need more medical resources, helping to plan health campaigns such as childhood vaccination drives or elderly care programmes.

EXERCISE 3.1

1. The ages (in years) of 12 learners in a Mathematics club are: 14,15,14,16,15,14,15,16,14,15,16,15. Draw a frequency distribution table and find the mean, mode and median age.

2. Using the grouped data below, find the mean mark:

Marks	0-9	10-19	20-29	30-39
Frequency	2	5	9	4

3. State the modal class in Question 2.

4. Explain the difference between a histogram and a bar graph.

5. Draw a frequency polygon for the data in Question 2 (use graph paper).

End of Topic Questions - Topic 3.1

1. The following are the scores (out of 20) of 9 learners in a quiz: 12, 15, 9, 18, 12, 14, 16, 12, 10. Find the mean, mode and median.

2. Find the median class for the grouped data: 1-10 (f=4), 11-20 (f=8), 21-30 (f=10), 31-40 (f=3).

3. Explain one advantage of representing data using a frequency polygon rather than raw numbers in a table.

Topic 3.2: Probability I

By the end of this topic, you should be able to:

- perform experiments involving probability and identify the range of probability;
- generate a probability space and determine mutually exclusive and independent events;
- apply the laws of probability, including tree diagrams.

3.2.1 Experimental Probability and Probability Space

KEY IDEA

Probability of an event, $P(E) = (\text{Number of favourable outcomes}) \div (\text{Total number of possible outcomes})$
 Probability always lies between 0 and 1: $0 \leq P(E) \leq 1$. $P(E) = 0$ means impossible; $P(E) = 1$ means certain.

The **probability space (sample space)** is the set of all possible outcomes of an experiment. For example, tossing a fair coin gives the sample space {Head, Tail}; rolling a fair die gives {1, 2, 3, 4, 5, 6}.

Worked Example 1

A fair die is rolled once. Find the probability of getting: (a) a 4 (b) an even number (c) a number greater than 6.

Solution:

Sample space = {1,2,3,4,5,6}, total outcomes = 6

(a) $P(4) = 1/6$

(b) Even numbers = {2,4,6} $\rightarrow P(\text{even}) = 3/6 = 1/2$

(c) No outcome is greater than 6 $\rightarrow P = 0$

3.2.2 Mutually Exclusive and Independent Events

KEY IDEA

Mutually exclusive events cannot happen at the same time (e.g. getting a head AND a tail on one coin toss). For mutually exclusive events A and B: $P(A \text{ or } B) = P(A) + P(B)$ **(Addition Law)**

Independent events do not affect each other's outcome (e.g. tossing a coin and rolling a die). For independent events A and B: $P(A \text{ and } B) = P(A) \times P(B)$ **(Multiplication Law)**

Worked Example 2

A bag contains 4 red balls and 6 blue balls. A ball is picked at random, replaced, then a second ball is picked. Find the probability that both balls picked are red.

Solution:

$P(\text{red}) = 4/10 = 2/5$ (each time, since the ball is replaced — independent events)

$P(\text{red and red}) = 2/5 \times 2/5 = 4/25$

3.2.3 Probability Tree Diagrams

A **tree diagram** shows all possible outcomes of two or more events as branches, with the probability of each outcome written along its branch. To find the probability of a combined outcome, multiply along the branches; to combine different paths, add the results.

Worked Example 3

A bag has 3 red and 2 white balls. Two balls are picked one after another without replacement. Draw a tree diagram and find the probability of picking one red and one white ball (in any order).

Solution:

$P(\text{Red then White}) = 3/5 \times 2/4 = 6/20 = 3/10$

$P(\text{White then Red}) = 2/5 \times 3/4 = 6/20 = 3/10$

$P(\text{one red and one white}) = 3/10 + 3/10 = 6/10 = 3/5$

THINK ABOUT IT

Some people believe that if a coin has landed on "heads" five times in a row, it is "due" to land on "tails" next. Using what you know about independent events, explain whether this belief is mathematically correct.

CASE STUDY: Weather Forecasting

Meteorologists often report "a 70% chance of rain tomorrow." This is a probability based on historical weather data and current atmospheric conditions. Farmers across Kenya use such probability forecasts to decide when to plant crops or harvest, showing how probability supports real economic decisions.

EXERCISE 3.2

1. A card is picked at random from a standard deck of 52 playing cards. Find the probability that it is a King.
2. A bag contains 5 green and 3 yellow sweets. One sweet is picked at random. Find $P(\text{green})$ and $P(\text{yellow})$.
3. Two fair coins are tossed together. List the sample space and find the probability of getting exactly one head.
4. A box has 4 defective and 16 good bulbs. Two bulbs are picked at random without replacement. Find the probability that both are defective.
5. Draw a tree diagram for tossing a fair coin twice, and use it to find the probability of getting two tails.

End of Topic Questions - Topic 3.2

1. A die is rolled once. Find the probability of getting a number less than 3 or an odd number (Hint: identify overlap carefully — these events are not mutually exclusive, list outcomes to be sure).
2. A bag contains 6 red and 4 blue balls. Two balls are picked one after another with replacement. Find the probability that both are blue.
3. Using a tree diagram, find the probability of getting at least one head when a fair coin is tossed twice.

STRAND WORKSHEETS

These worksheets mix questions from across each strand for extra revision practice. Complete each worksheet in your exercise book, then check your answers in the Answer Key chapter at the end of this book.

Worksheet 1: Numbers and Algebra

Name: _____ Class: _____ Date: _____

1. State whether 91 is a prime or composite number. Show your working.
2. Find the reciprocal of 0.16.
3. Simplify $5^3 \times 5^{-1}$.
4. Use logarithms to evaluate 12.5×8.4 .
5. Factorise: $x^2 + x - 6$.
6. Solve the quadratic equation: $x^2 - 4x + 3 = 0$.

Worksheet 2: Measurements and Geometry

Name: _____ Class: _____ Date: _____

1. Two similar shapes have a linear scale factor of 4. If the area of the smaller shape is 10 cm^2 , find the area of the larger shape.
2. Reflect the point $(3, -5)$ in the x-axis.
3. Rotate the point $(2, 6)$ through 180° about the origin.
4. Find the value of $\sin 60^\circ \times \cos 30^\circ$.
5. Find the area of a right-angled isosceles triangle with two equal sides of 5 cm (included angle 90°).
6. Find the area of a semicircle of radius 7 cm. (Take $\pi = 22/7$)
7. Find the volume of a cylinder of radius 3 cm and height 10 cm. (Take $\pi = 3.142$)
8. Find the magnitude of the vector $(6, 8)$.

Worksheet 3: Statistics and Probability

Name: _____ Class: _____ Date: _____

1. Find the mean of: 4, 8, 6, 10, 2.
2. Find the mode of: 3, 5, 5, 7, 5, 9.
3. A fair coin is tossed once. Find the probability of getting a tail.
4. A fair die is rolled once. Find the probability of getting a number greater than 4.
5. State, with a reason, whether picking a red card and picking a black card from a standard deck (in a single draw) are mutually exclusive events.

ANSWER KEY

This chapter contains answers to all Exercises, End of Topic Questions, and Worksheets found in this book. Use it to check your work after attempting each question on your own.

STRAND 1: NUMBERS AND ALGEBRA

Topic 1.1: Real Numbers

Exercise 1.1

- 21 – odd, composite (3×7); 32 – even, composite; 41 – odd, prime; 57 – odd, composite (3×19); 64 – even, composite.
- 0.25 rational; $\sqrt{5}$ irrational; 7 rational; $\sqrt{16} = 4$ rational; 1.101001000... irrational (non-recurring).
- (a) $1/12 \approx 0.083$ (b) 4 (c) $1\frac{2}{5} = 7/5$, reciprocal = $5/7 \approx 0.714$ (d) 8
- $x + 1/x = 2 \Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x-1)^2 = 0 \Rightarrow x = 1$
- $1 \div 3.6 \approx 0.278$
- Every whole number greater than 1 has factors 1 and itself; if these are its only factors, it is prime; if it has more factors, it is composite — there is no other possibility.

End of Topic Questions 1.1

- 41, 43, 47, 53, 59
- $0.4545... = 45/99 = 5/11$
- Reciprocal of $5\frac{1}{3}$ ($=16/3$) is $3/16$; product = $16/3 \times 3/16 = 1$
- $1 \div (3/8) = 8/3 \approx 2.67$ hours
- 1 is neither prime nor composite — it has only one factor (itself).

Topic 1.2: Indices and Logarithms

Exercise 1.2

- (a) $3^6 = 729$ (b) $7^6 = 117\,649$ (c) $2^6 = 64$ (d) $6^{-1} = 1/6$
- $\log_{10} 1000 = 3$
- $10^5 = 100\,000$
- $34.5 \times 2.16 \approx 74.5$
- $456.7 \div 12.3 \approx 37.1$
- $\sqrt{28.9} \approx 5.38$
- $x + 3 = 9 \Rightarrow x = 6$

End of Topic Questions 1.2

- $5^2 = 25$
- $\log 6 = \log 2 + \log 3 = 0.3010 + 0.4771 = 0.7781$
- $(23.6 \times 1.98) \div 4.05 \approx 11.5$
- $3^x = 3^4 \Rightarrow x = 4$
- Number of bacteria after n hours = $2 \times 2^n = 2^{n+1}$; after 5 hours = $2^6 = 64$

Topic 1.3: Quadratic Expressions and Equations

Exercise 1.3

- (a) $x^2 + 8x + 16$ (b) $4x^2 - 12x + 9$ (c) $x^2 - 36$
- (a) $(x+3)(x-3)$ (b) $(x+3)(x+4)$ (c) $(x-5)(x+2)$
- (a) $x = 3$ or 4 (b) $x = -5$ or 3 (c) $x = -\frac{1}{2}$ or 3
- Width = 5 cm, Length = 9 cm
- $x^2 - x - 12 = 0$

End of Topic Questions 1.3

- $98^2 = (100-2)^2 = 10\,000 - 400 + 4 = 9604$
- $3(x-2)(x+2)$
- $x = 3$ or -2
- $t = 0$ or $t = 4$

5. The integers are 9 and 10.

STRAND 2: MEASUREMENTS AND GEOMETRY

Topic 2.1: Similarity and Enlargement

Exercise 2.1

1. L.S.F = 3; ratio of areas = 1 : 9
2. L.S.F = 3, A.S.F = 9, V.S.F = 27
3. Actual length = $8 \times 50\,000\text{ cm} = 400\,000\text{ cm} = 4\text{ km}$
4. V.S.F = $216/64 = 27/8$, so L.S.F (height ratio) = 3 : 2
5. Join corresponding points on the object and image with straight lines; extend these lines — the point where they all meet is the centre of enlargement.

End of Topic Questions 2.1

1. L.S.F = $\sqrt{(18/50)} = 3/5$, so sides are in ratio 3 : 5
2. V.S.F = $2^3 = 8$; Volume of larger solid = $27 \times 8 = 216\text{ cm}^3$
3. L.S.F = $27/9 = 3$; new width = $6 \times 3 = 18\text{ cm}$; A.S.F = 9

Topic 2.2: Reflection and Congruence

Exercise 2.2

1. A'(1, -1), B'(4, -1), C'(4, -3)
2. $(5, -2) \rightarrow (-2, 5)$
3. Mirror line: $y = 3$
4. RHS (Right angle, Hypotenuse, Side)
5. A square has 4 lines of symmetry.

End of Topic Questions 2.2

1. Reflection in y-axis: (3,4); Reflection in x-axis: (-3, -4)
2. An isosceles triangle has 1 line of symmetry.
3. SSS (Side, Side, Side)

Topic 2.3: Rotation

Exercise 2.3

1. (2, -5)
2. (-1, -4)
3. (a) 3 (b) 6 (c) 2
4. 90° anticlockwise
5. Infinite order of rotational symmetry.

End of Topic Questions 2.3

1. (-5, 0)
2. Order of rotational symmetry = 8
3. Reflection flips a shape over a mirror line to form a mirror image (e.g. your reflection in a mirror); rotation turns a shape about a fixed point through an angle (e.g. the hands of a clock).

Topic 2.4: Trigonometry I

Exercise 2.4

1. $\sin 42^\circ \approx 0.6691$, $\cos 42^\circ \approx 0.7431$, $\tan 42^\circ \approx 0.9004$
2. Height = $12 \times \tan 35^\circ \approx 8.40\text{ m}$
3. $\sin 30^\circ + \cos 60^\circ = 0.5 + 0.5 = 1$

4. $d = 15 \div \tan 20^\circ \approx 41.2 \text{ m}$
5. $\sin 50^\circ = \cos(90^\circ - 50^\circ) = \cos 40^\circ$ (verified using the complementary angle rule)

End of Topic Questions 2.4

1. $L = 2 \div \cos 70^\circ \approx 5.85 \text{ m}$
2. $\text{Height} = 20 \times \tan 40^\circ \approx 16.78 \text{ m}$
3. $\tan 45^\circ + \sin 30^\circ \times \cos 60^\circ = 1 + (0.5 \times 0.5) = 1.25$

Topic 2.5: Area of Polygons

Exercise 2.5

1. $\text{Area} = \frac{1}{2} \times 6 \times 9 \times \sin 65^\circ \approx 24.5 \text{ cm}^2$
2. $s = 9$; $\text{Area} = \sqrt{(9 \times 4 \times 3 \times 2)} = \sqrt{216} \approx 14.7 \text{ cm}^2$
3. $\text{Area} = \frac{1}{2}(8+12) \times 5 = 50 \text{ cm}^2$
4. $\text{Area} = 6 \times (\sqrt{3}/4) \times 4^2 = 24\sqrt{3} \approx 41.6 \text{ cm}^2$
5. $\text{Area} = \frac{1}{2} \times 10 \times 14 = 70 \text{ cm}^2$

End of Topic Questions 2.5

1. $\text{Area} = 9 \times 6 \times \sin 75^\circ \approx 52.2 \text{ cm}^2$
2. $s = 28.5$; $\text{Area} = \sqrt{(28.5 \times 13.5 \times 10.5 \times 4.5)} \approx 134.8 \text{ m}^2$
3. $\text{Area} \approx 43.5 \text{ cm}^2$

Topic 2.6: Area of a Part of a Circle

Exercise 2.6

1. $\text{Area} = (90/360) \times 22/7 \times 21^2 = 346.5 \text{ cm}^2$
2. $\text{Area} = 3.142 \times (10^2 - 6^2) = 201.09 \text{ cm}^2$
3. $\text{Sector area} \approx 55.86 \text{ cm}^2$; $\text{triangle area} \approx 31.51 \text{ cm}^2$; $\text{segment area} \approx 24.35 \text{ cm}^2$
4. $\text{Area} = (45/360) \times 3.142 \times (12^2 - 8^2) = 31.42 \text{ cm}^2$
5. Divide the overlapping region into two circular segments (one from each circle), calculate each segment's area using the sector-minus-triangle method, and add the two segment areas together.

End of Topic Questions 2.6

1. $\text{Area} \approx 144.3 \text{ cm}^2$
2. $\text{Area} \approx 1675.7 \text{ cm}^2$
3. $\text{Area} \approx 16.7 \text{ cm}^2$

Topic 2.7: Surface Area and Volume of Solids

Exercise 2.7

1. $\text{Volume} = 24 \times 15 = 360 \text{ cm}^3$
2. $\text{Volume} = \frac{43}{7} \times 22/7 \times 7^3 \approx 1437.3 \text{ cm}^3$
3. $S.A = \pi r l + \pi r^2 = \pi(65+25) = 90\pi \approx 282.8 \text{ cm}^2$
4. $\text{Full cone volume} = 324\pi$; $\text{small cone volume} = 12\pi$; $\text{frustum volume} = 312\pi \approx 980.3 \text{ cm}^3$
5. $\text{Cube volume} = 216 \text{ cm}^3$; $\text{pyramid volume} = 48 \text{ cm}^3$; $\text{total} = 264 \text{ cm}^3$

End of Topic Questions 2.7

1. $S.A = 4\pi(10.5)^2 \approx 1385.9 \text{ cm}^2$
2. $\text{Volume} = \pi(2.1)^2(3) \approx 41.57 \text{ m}^3 = 41\,570 \text{ litres}$
3. $\text{Volume} = \frac{1}{3} \times 8^2 \times 15 = 320 \text{ cm}^3$

Topic 2.8: Vectors I

Exercise 2.8

1. $\vec{PQ} = (4,6)$; $|\vec{PQ}| = \sqrt{52} \approx 7.21$
2. $\vec{a} + \vec{b} = (2,6)$; $\vec{a} - \vec{b} = (4,2)$
3. Midpoint = (2, 1)
4. Magnitude = $\sqrt{169} = 13$
5. $3\vec{u} = (6, -9)$

End of Topic Questions 2.8

1. Translation vector = (5, -4)
2. $2\vec{a} + \vec{b} = (9,1)$; magnitude = $\sqrt{82} \approx 9.06$
3. $\vec{AB} = (4,8)$; distance AB = $\sqrt{80} \approx 8.94$

Topic 2.9: Linear Motion

Exercise 2.9

1. Speed = $15 \div 0.75 = 20$ km/h
2. Acceleration = $(0 - 30)/6 = -5$ m/s² (a deceleration)
3. Time = $300 \div (50 + 70) = 2.5$ hours
4. Total distance = area under graph = $(\frac{1}{2} \times 10 \times 20) + (20 \times 15) = 100 + 300 = 400$ m
5. Distance = total length of the path walked (the perimeter of the field, a positive value); displacement = zero, since the learner returns to the starting point.

End of Topic Questions 2.9

1. Speed = $240 \div 3 = 80$ km/h
2. Relative speed = $110 - 90 = 20$ km/h
3. $v = 0 + (2 \times 8) = 16$ m/s; distance = $\frac{1}{2}(2)(8^2) = 64$ m

STRAND 3: STATISTICS AND PROBABILITY

Topic 3.1: Statistics I

Exercise 3.1

1. Frequencies: 14→4, 15→5, 16→3. Mean = $179/12 \approx 14.92$; Mode = 15; Median = 15
2. Mean = $440/20 = 22$
3. Modal class = 20–29
4. A histogram is used for continuous grouped data with bars touching (no gaps); a bar graph is used for discrete/categorical data with gaps between bars.
5. (Graphical answer — plot the midpoints of each bar's top against frequency and join with straight lines.)

End of Topic Questions 3.1

1. Mean = $118/9 \approx 13.11$; Mode = 12; Median = 12
2. Median class = 21–30 (13th value falls here, using cumulative frequency 4, 12, 22, 25)
3. A frequency polygon shows the overall shape/trend of a distribution clearly and allows easy comparison of two or more distributions plotted on the same axes.

Topic 3.2: Probability I

Exercise 3.2

1. $P(\text{King}) = 4/52 = 1/13$
2. $P(\text{green}) = 5/8$; $P(\text{yellow}) = 3/8$
3. Sample space: {HH, HT, TH, TT}; $P(\text{exactly one head}) = 2/4 = 1/2$
4. $P(\text{both defective}) = (4/20) \times (3/19) = 3/95$
5. $P(\text{two tails}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

End of Topic Questions 3.2

1. "Less than 3" = {1,2}; "odd" = {1,3,5}; union = {1,2,3,5}; $P = 4/6 = 2/3$
2. $P(\text{both blue}) = (4/10) \times (4/10) = 4/25$
3. $P(\text{at least one head}) = 1 - P(\text{no heads}) = 1 - \frac{1}{4} = \frac{3}{4}$

WORKSHEET ANSWERS

Worksheet 1: Numbers and Algebra

- 1. $91 = 7 \times 13 \rightarrow$ composite
- 2. Reciprocal of $0.16 = 1/0.16 = 6.25$
- 3. $5^3 \times 5^{-1} = 5^2 = 25$
- 4. $12.5 \times 8.4 = 105$
- 5. $x^2+x-6 = (x+3)(x-2)$
- 6. $x^2-4x+3 = 0 \Rightarrow (x-1)(x-3)=0 \Rightarrow x = 1$ or 3

Worksheet 2: Measurements and Geometry

- 1. Area of larger shape $= 10 \times 4^2 = 160 \text{ cm}^2$
- 2. $(3, -5)$ reflected in x-axis $\rightarrow (3, 5)$
- 3. $(2, 6)$ rotated 180° about origin $\rightarrow (-2, -6)$
- 4. $\sin 60^\circ \times \cos 30^\circ = (\sqrt{3}/2) \times (\sqrt{3}/2) = 3/4 = 0.75$
- 5. Area $= \frac{1}{2} \times 5 \times 5 \times \sin 90^\circ = 12.5 \text{ cm}^2$
- 6. Area of semicircle $= \frac{1}{2} \times 22/7 \times 7^2 = 77 \text{ cm}^2$
- 7. Volume $= 3.142 \times 3^2 \times 10 = 282.78 \text{ cm}^3$
- 8. Magnitude of $(6,8) = \sqrt{(36+64)} = \sqrt{100} = 10$

Worksheet 3: Statistics and Probability

- 1. Mean $= (4+8+6+10+2)/5 = 30/5 = 6$
- 2. Mode $= 5$
- 3. $P(\text{tail}) = \frac{1}{2}$
- 4. $P(\text{number} > 4) = \{5,6\}/6 = 2/6 = \frac{1}{3}$
- 5. Yes, they are mutually exclusive — a single card cannot be both red and black at the same time.